

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460227.8569 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.062
Transit depth (Rp/Rs)^2	1.59 +/- 1.55 [%]
Orbital Inclination (inc)	84.01 +/- 1.9
Airmass coefficient 1 (a1)	1.43 +/- 1.12
Airmass coefficient 2 (a2)	-0.1 +/- 0.46
Scatter in the residuals of the lightcurve fit is	93.76 %
Best Comparison Star	#3 - [278, 293]
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.025 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.126 $\pm$ 0.062 vs 0.1592 (deviation 0.42 σ)
Duration fit vs published	0.025 d vs 0.0946 d (deviation 1.76 σ)
Mid-time O-C	1.8 min
Depth SNR	1.6
Residual scatter	93.76 %

Light curve

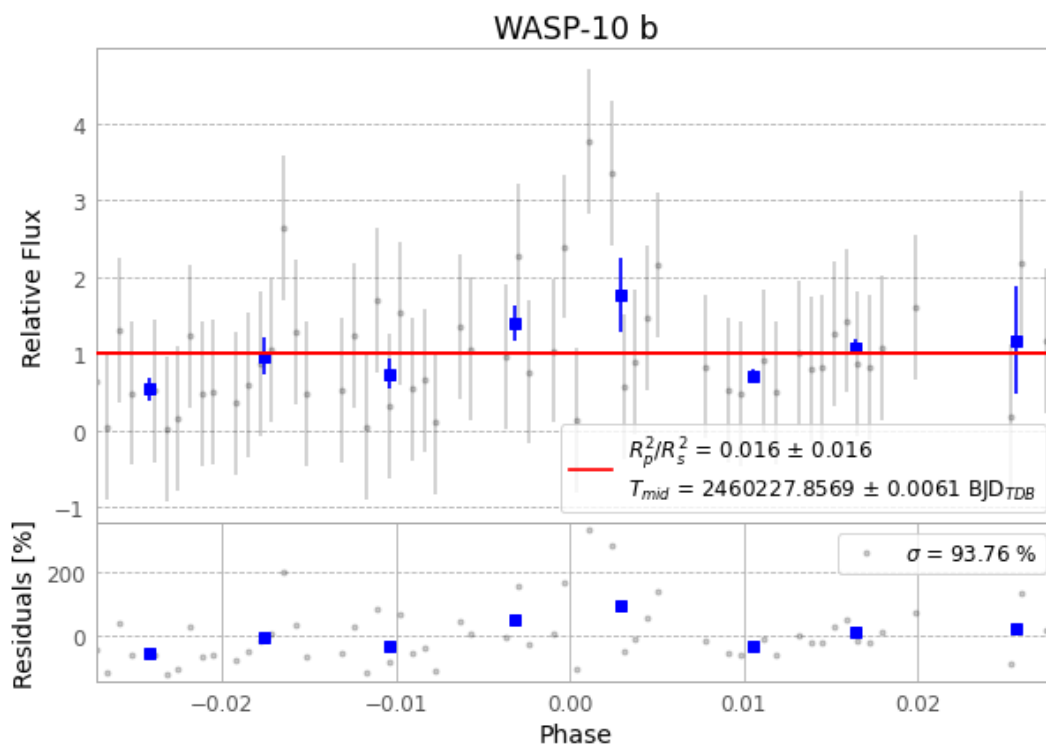


Figure 1 — FinalLightCurve_WASP-10 b_2023-10-09.png (SHA-256 737fa81bd68ed1a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [145, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02317e3b078e4b59...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit aa46c9a

Data provenance

85 FITS frames (56 MB), WASP-10231010062115.FITS → WASP-10231010103316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231010.log (5a16c298c586...), FinalLightCurve_WASP-10 b_2023-10-09.png (737fa81bd68e...), FinalLightCurve_WASP-10 b_2023-10-09.csv (6c221263f4e4...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:38Z.